Nongeometric primal convergence of an inertial hinge algorithm

Abstract

We give a finite-dimensional counterexample to a universal eventual linear rate for the primal iterates of the projected inertial dual-hinge algorithm. The signed data are linearly separable and the two initial dual states are equal and nonzero. We treat both the allowed maximal step and the fixed strict step $\gamma = 1-10^{-30}$ when the Lipschitz constant is one. The minimum-norm primal separator is unique, whereas the dual solution set is a triangle. For each of these two steps, a compactness argument selects one fixed initialization whose trajectory has infinitely many isolated face contacts separated by geometrically increasing waiting times. At actual integer iterates, the primal error has a polynomial subsequence lower bound and therefore satisfies no eventual R-linear estimate. The construction uses the unchanged algorithm and a prescribed decreasing box schedule. At the strict step, the proof retains both range-memory states and controls them relative to the shrinking contact scale at every return. At the maximal step, the mechanism is Nesterov-extrapolated alternating projection between an affine space and an orthant. The result does not cover every strict step or the zero-vector initialization. The lower-bound exponent is deliberately conservative.

Keywords: inertial forward backward method, hinge loss, alternating projections, nonunique minimizers

2020 Mathematics Subject Classification. Primary 90C25; Secondary 65K10, 49M37.

1 The algorithm and the question

Let $Z \in \mathbb{R}^{n\times d}$ have rows $y_i x_i^T$, where $y_i \in \{-1, 1\}$, and suppose the data are strictly linearly separable. The minimum-norm separator is the unique solution of

$$\min\left\{\frac{1}{2}\|w\|^2 : Zw \geq \mathbf{1}\right\}.$$

Write $Q = ZZ^T$ and $L = \|Q\|$. Algorithm 2 of [1] has the form

$$\begin{aligned} q_k &= u_k + \frac{k}{k+\alpha}(u_k - u_{k-1}),\\ u_{k+1} &= P_{[-1/\lambda_k,0]^n}\big((I-\gamma Q)q_k - \gamma\mathbf{1}\big),\\ w_k &= -Z^T u_k, \qquad k \geq 1, \end{aligned} \tag{1}$$

where $u_0 = u_1 \in [-1/\lambda_0, 0]^n$, $\alpha \geq 3$, $0 < \gamma \leq 1/L$, and $\lambda_k > 0$ decreases to zero. Theorem 5.2 of [1] gives convergence of w_k and an $O(k^{-1})$ distance bound. Remark 5.1 asks whether a linear rate can be proved for the inertial variant. Our result concerns that question; it does not contradict the stated sublinear theorem.

For clarity, eventual R-linear convergence here means the existence of $C < \infty$, $\rho \in (0,1)$ and K such that

$$\|w_k - w_*\| \le C\rho^k \qquad (k \ge K). \tag{2}$$

The example below violates (2) at $\gamma = 1/L$ and at the single fixed strict step $\gamma = 1-10^{-30}$ with $L = 1$. It uses zero initial velocity, $u_0 = u_1$, but not $u_0 = 0$. The strict-step value is fixed before the initialization is chosen; it does not depend on a finite time horizon.

Error-bound analyses of inertial forward–backward methods already give strong polynomial conclusions at strict steps [2]; the lower subsequence proved here does not contradict those polynomial conclusions. Infinite face switching is also not new: the stationary generalized alternating-projection example in [3, Example 6.2] still converges geometrically. The relevant feature here is the growing separation of contacts together with a lower bound on the visible primal error. We make no claim that the conservative exponent below is optimal.

2 An explicit family of separable instances

Set $n = d = 11$, $a = 1/\sqrt{2}$, $b = 1/\sqrt{110}$, and let $B \in \mathbb{R}^{11\times 2}$ have rows

$$(a,b), \quad (-a,b), \quad \underbrace{(0,b),\dots,(0,b)}_{8 \text{ rows}}, \quad (0,-10b). \tag{3}$$

Define

$$Q = I - BB^T, \quad Z = Q, \quad \gamma = 1-\delta, \quad \alpha = 3, \quad \lambda_k = \frac{1}{1000+k}, \quad \delta \in \{0, 10^{-30}\}. \tag{4}$$

All entries of Q are rational. Directly, $B^TB = I_2$ and $B^T\mathbf{1} = 0$, so Q is an orthogonal projector of rank nine, $Q\mathbf{1} = \mathbf{1}$, and $ZZ^T = Q$ has norm one. The rows can be realized as signed data with either one or both labels: choose any signs y_i and set $x_i^T = y_i Z_i$.

Fix $\varepsilon = 10^{-12}$. For $c \in [-2, 1/2]$ put

$$\begin{aligned} Y_0 &= \frac{22}{9} + \frac{38}{9}\delta - \frac{11}{9}c\varepsilon, \quad t_0 = \big(\sqrt{2}\varepsilon, \sqrt{110}(Y_0 - 1)\big), \\ v_0 &= v_1 = 20\,\mathbf{1} + Bt_0, \qquad u_0 = u_1 = -v_0. \end{aligned} \tag{5}$$

The parameter c will be chosen once, by a connectedness argument, and will then remain fixed for the entire orbit.

Theorem 1. *For each $\delta \in \{0, 10^{-30}\}$, the data and parameters (3)–(4) admit a fixed $c_* = c_*(\delta) \in [-2, 1/2]$ such that the legal initialization (5) produces $w_k \to w_* = \mathbf{1}$ and an unbounded sequence of integer indices m_ℓ with*

$$\|w_{m_\ell} - \mathbf{1}\| \ge C m_\ell^{-q}, \qquad q = \frac{\log 200}{\log 1.4}, \qquad C > 0. \tag{6}$$

Consequently, this orbit satisfies no eventual R-linear estimate. In both cases the dual orbit converges to the degenerate optimum $u_* = -11e_{11}$.

The energy estimate below applies to both steps. We first give the endpoint proof, where the null variables close exactly. Sections 9–12 then establish the fixed strict-step assertion with the complete range state retained.

The positive multipliers $v = -u$ make the geometry transparent. For $v \geq 0$,

$$F(v) := \frac{1}{2} v^T Q v - \mathbf{1}^T v = \frac{1}{2} \|Qv - \mathbf{1}\|^2 - \frac{11}{2}. \tag{7}$$

Its minimizers are $\{\mathbf{1} + Bt : \mathbf{1} + Bt \geq 0\}$. In the coordinates

$$X = at_x, \quad Y = 1 + bt_y, \quad x = Y + X, \quad y = Y - X, \tag{8}$$

the constraints are

$$x \geq 0, \qquad y \geq 0, \qquad x + y \leq 11/5;$$

the eight remaining inequalities $(x + y)/2 \geq 0$ are redundant. Thus the dual solution set is a genuine triangle. Its lower apex is $t_* = (0, -\sqrt{110})$. The example has strictly positive dual optima, including $v = \mathbf{1}$; the limiting dual point of the exceptional orbit is degenerate, rather than every dual optimum being degenerate.

If $Zw \geq \mathbf{1}$, then $\mathbf{1}^T w = \mathbf{1}^T Z w \geq 11$. Cauchy–Schwarz gives $\|w\| \geq \sqrt{11}$, with equality only at $w = \mathbf{1}$. This proves strict separability and the asserted unique minimum-norm separator.

3 A direct energy bound and the exact reduction

We record a standard estimate with its short proof to avoid assuming boundedness when removing the boxes.

Lemma 1. *Let $F = f + g$, where f is convex with one-Lipschitz gradient and g is proper closed convex. Suppose $x_{k+1} = \operatorname{prox}_{\gamma g}(q_k - \gamma \nabla f(q_k))$, $0 < \gamma \leq 1$, $q_k = x_k + k(x_k - x_{k-1})/(k+3)$, and x_* minimizes F. Put*

$$\tau_k = (k+2)/2, \qquad z_k = \tau_k x_k - (\tau_k - 1) x_{k-1}.$$

Then

$$\mathcal{E}_k = \gamma \tau_k^2 (F(x_k) - F(x_*)) + \tfrac{1}{2} \|z_k - x_*\|^2 \tag{9}$$

is nonincreasing.

Proof. The quadratic upper model and the proximal variational inequality give, for every feasible z,

$$2\gamma(F(x_{k+1}) - F(z)) \leq \|z - q_k\|^2 - \|z - x_{k+1}\|^2 - (1 - \gamma) \|x_{k+1} - q_k\|^2 .$$

Take $z = (1 - 1/\tau_{k+1})x_k + x_*/\tau_{k+1}$, use convexity, and multiply by τ_{k+1}^2. The two squares become $\|z_k - x_*\|^2$ and $\|z_{k+1} - x_*\|^2$. Since

$$\frac{\tau_k - 1}{\tau_{k+1}} = \frac{k}{k+3}, \qquad \tau_{k+1}^2 - \tau_{k+1} = \tau_k^2 - \tfrac{1}{4},$$

the claimed estimate follows. ∎

Apply this argument directly to the actual changing boxes. At step k the positive box is $C_k = [0, 1000 + k]^{11}$. The preceding iterate v_k and the point $\mathbf{1}$ both belong to C_k, because the boxes expand. Hence their convex combination used in the proof is feasible in the current projection inequality. The same proof, with F in (7) and the current box indicator, gives

$$E_k = \gamma\tau_k^2(F(v_k) - F(\mathbf{1})) + \tfrac{1}{2}\|z_k - \mathbf{1}\|^2, \qquad E_{k+1} \le E_k. \tag{10}$$

This statement does not presume inactivity of the upper box. At the equal initialization,

$$F(v_1) - F(\mathbf{1}) = 3971/2, \quad z_1 = v_1, \qquad \|v_1 - \mathbf{1}\|^2 = 3971 + 2\varepsilon^2 + 110(Y_0 - 1)^2 < 4202.$$

Since $\gamma \le 1$, we have $2E_1 < 13136.75 < 115^2$. The identity

$$v_k - \mathbf{1} = \frac{z_k - \mathbf{1}}{\tau_k} + \left(1 - \frac{1}{\tau_k}\right)(v_{k-1} - \mathbf{1})$$

proves $\|v_k - \mathbf{1}\| < 115$ and $\|v_k\| < 119$. Every output coordinate is therefore strictly below its upper bound, so the upper boxes are inactive. The initialization coordinates are $19 + Y_0 \pm \varepsilon$, eight copies of $19 + Y_0$, and $30 - 10Y_0$; all lie strictly between five and twenty-two. The initialization and the entire schedule are legal for both values of δ. Equations (7) and (10) also give $\|w_k - \mathbf{1}\| = O(k^{-1})$ directly.

For the remainder of this section and through Section 8, set $\delta = 0$. This endpoint case explains the contact mechanism before range memory is restored.

With $t_k = B^T v_k$, the exact reduced iteration is

$$t_{k+1} = B^T\left[\mathbf{1} + B\big(t_k + \beta_k(t_k - t_{k-1})\big)\right]_+, \qquad \beta_k = \frac{k}{k+3}. \tag{11}$$

For $q = t_k + \beta_k(t_k - t_{k-1})$ and $\xi = [-(\mathbf{1} + Bq)]_+$, we have

$$w_{k+1} - \mathbf{1} = Q\xi, \qquad \|Q\xi\| \ge \frac{\|\xi\|_1}{\sqrt{11}}. \tag{12}$$

Here $\xi \ge 0$ and $\mathbf{1}^T Q\xi = \mathbf{1}^T\xi$, so the last inequality is just Cauchy–Schwarz. It will ensure that a contact produces a visible primal error.

At the maximal step, the affine map $q \mapsto \mathbf{1} + BB^T q$ is projection onto $\mathbf{1}+\operatorname{range} B$. Thus the positive-multiplier iteration is exactly Nesterov-extrapolated alternating projection from that affine space onto the positive orthant.

4 The exact free clock and the cap launch

Define for real $r > 0$

$$b(r) = \frac{1}{r(r+1)(r+2)}, \qquad H(r) = \frac{1}{2(r+1)(r+2)}. \tag{13}$$

On a free segment, $p = (t_k - t_{k-1})/b(k)$ is constant, since $b(k+1)/b(k) = \beta_k$. Moreover $H(k-1) - H(k) = b(k)$, and

$$t_k = R - pH(k). \tag{14}$$

Interpolate genuinely free updates affinely in H. This is an exact discrete identity, not a continuous-time approximation.

At step two only the cap clips. Direct substitution into (11) yields

$$X_2 = \varepsilon, \quad Y_2 = 1 + Y_0/11 = 11/9 - c\varepsilon/9.$$

The next free proposal is $Y_3 = 11/15 + c\varepsilon/3$, strictly inside the triangle. Thereafter, until the first lower contact,

$$X_k = \varepsilon, \qquad Y_k = c\varepsilon + P_0 H(k), \qquad P_0 = (88 - 80c\varepsilon)/3 > 29. \tag{15}$$

The first lower wall is $y = Y - X$. Its ideal incoming clock is

$$H_1 = (1-c)\varepsilon/P_0. \tag{16}$$

If j is the last free nonnegative pre-index, then $H(j+1) \le H_1 \le H(j)$, with

$$2 \cdot 10^6 < j < 6 \cdot 10^6. \tag{17}$$

These inequalities follow by inserting $c \in [-2, 1/2]$ in (16). Equality of a crossing with an integer is handled by the zero-impulse convention below.

It is useful to exclude cap recontact globally. The reduced map is gradient descent with inertia for

$$f(t) = \tfrac{1}{2} \left\| [\mathbf{1} + Bt]_- \right\|^2,$$

a convex function with gradient Lipschitz constant at most one. At the pre-index j, $f(t_j) = 0$ and

$$\tau_j t_j - (\tau_j - 1) t_{j-1} - t_* = (\sqrt{2}\varepsilon, \sqrt{110}c\varepsilon).$$

Its norm is less than 22ε. The direct free formulas also give $\|t_j - t_*\| < 22\varepsilon$. Lemma 1 and the convex-combination identity imply $\|t_k - t_*\| < 22\varepsilon$ for $k \ge j$. Every later predictor is within 66ε of the apex; its cap coordinate is strictly positive. Thus there is exactly one cap clipping, at step two.

5 A phase uniform contact estimate

The Gram matrix of the lower-wall row normals is

$$\frac{1}{55}\begin{pmatrix} 28 & -27 \\ -27 & 28 \end{pmatrix}.$$

Put $d_0 = 27/55$ and $\vartheta = 27/28$. Suppose a target wall z is approached with scaled momentum $-P$ at a pre-index $j \geq 10^6$. Set $A = Pb(j+1)$ and $a_0 = z_j/A \in [0,1]$. In this section a_0 is an entry phase, unrelated to the coefficient a in (3).

If only that wall clips, its normalized recurrence is

$$u_0 = a_0, \quad u_1 = d_0(a_0 - 1), \quad u_n = \psi\big((1+\beta_{j+n-1})u_{n-1} - \beta_{j+n-1}u_{n-2}\big), \tag{18}$$

where $\psi(s) = d_0 s$ for $s < 0$ and $\psi(s) = s$ for $s \geq 0$. The first prediction is exactly $a_0 - 1$, because the previous free increment is $A(j+3)/j$. We first establish a forced single-wall estimate; isolation from the other coordinates is proved afterward.

Lemma 2. *The recurrence* (18) *has three or four negative steps, followed by a free release. Its scaled restitution e, the positive outgoing scaled speed divided by P, satisfies*

$$.23 < e < .26. \tag{19}$$

Zero predictions may be assigned to either neighboring phase description; they have zero impulse and do not change the trajectory.

Proof. Freeze the coefficients at $\beta = 1$. Exact expansion gives the second, third and fourth predictors

$$\begin{aligned} p_2 &= -(54 + a_0)/55, & u_2 &= -(1458 + 27a_0)/3025, \\ p_3 &= -(1431 + 1539a_0)/3025, & u_3 &= -(38637 + 41553a_0)/166375, \\ p_4 &= (2916 - 81621a_0)/166375. \end{aligned}$$

The transition phase is $a_* = 108/3023$. In the branch where the fourth step clips,

$$u_4 = \frac{78732 - 2203767a_0}{9150625}, \qquad p_5 = \frac{2282499 - 2122119a_0}{9150625}.$$

The second and third predictors are below $-.98$ and $-.47$, respectively, and the last displayed fifth predictor is above .017. The outgoing velocities on the two branches are

$$V_4 = \frac{41553 - 40068a_0}{166375}, \qquad V_5 = \frac{2203767 + 81648a_0}{9150625}. \tag{20}$$

They agree at a_* and belong to $(.241, .250)$ on their respective phase intervals.

For the actual coefficients, $|\beta_i - 1| \leq 3/j$. The relevant frozen successive differences have magnitude at most one, and the scalar clipping map is one-Lipschitz. Thus state errors satisfy

$$E_n \leq 2E_{n-1} + E_{n-2} + 3/j, \qquad E_0 = E_1 = 0.$$

In particular $E_2 \le 3/j$, $E_3 \le 9/j$, the fourth-predictor error is at most $24/j$, and the fifth-predictor error in the fourth-clipped branch is at most $60/j$. The second and third signs remain negative; the fifth remains positive when needed.

Release at step four requires $a_0 \le a_* + 49/j$; release at step five requires $a_0 \ge a_* - 49/j$. Substitution in (20), together with the preceding error bounds, gives the common enclosure

$$.241 - 85/j < V < .250 + 84/j$$

for the released physical velocity divided by A. If release is at step $N \in \{4, 5\}$, then

$$e = V\frac{b(j+1)}{b(j+N)}, \qquad 1 \le \frac{b(j+1)}{b(j+N)} < 1 + 13/j.$$

For $j \ge 10^6$ these bounds imply (19). They also exclude a vanishing outgoing velocity. If a first or fourth prediction is exactly zero, its disappearing correction is zero; shifting the entry or release convention leaves the same ordinary states. ∎

6 An invariant contact block

Define a section at the outgoing zero crossing of the just-contacted wall on a genuinely free release segment. Orient its wall coordinates as $w = 0$, $z = y > 0$. Write the scaled wall momenta and clock as

$$p_w = sP, \qquad p_z = -P, \qquad P > 0, \qquad H = H(\kappa), \qquad y = \chi HP. \tag{21}$$

Consider the block

$$\kappa \ge 10^6, \qquad .1 \le s \le .32, \qquad .5 \le \chi \le .96, \qquad 0 < y < 2.2. \tag{22}$$

The next section interchanges the wall orientation.

Lemma 3. *From every realized section in* (22)*, the next opposite-wall contact is isolated and is followed by a genuine free outgoing crossing. No other lower, middle or cap coordinate clips during that return. For its incoming pre-index $j \ge \kappa$, let e be the restitution in Lemma 2, and put $r = (1 + e)\vartheta - s$. Then*

$$P' = rP, \quad s' = e/r, \quad .866 < r < 1.115, \quad .206 < s' < .301. \tag{23}$$

If $\overline{H} = H(1 - \chi)$ and $\delta_H = H' - \overline{H}$, then

$$|\delta_H| \le 5b(j+1), \qquad D' = s\chi HP + (\vartheta - s)P\delta_H, \tag{24}$$

where D' is the opposite distance at the next section. In particular,

$$\left|\chi' - \frac{s}{r}\frac{\chi}{1-\chi}\right| < 100/j, \qquad 0 < D' < .321y < 2.2. \tag{25}$$

The lower face $\chi = .5$ maps strictly below .5, and the upper face $\chi = .96$ maps strictly above .96.

Proof. During the free flight the target reaches zero at clock $\overline{H}$. Since $\chi \geq 1/2$, $H(\kappa+1) > H(\kappa)/2 \geq \overline{H}$, so the encounter occurs more than one ordinary step after the outgoing section. Its last nonnegative pre-index satisfies $j \geq \kappa$.

The crucial tangent coordinate is

$$h = w + \vartheta z. \tag{26}$$

Target-wall clipping leaves it exactly unchanged: the off-diagonal normal Gram entry is $-\vartheta$ times the diagonal entry. Its scaled momentum is therefore $(s-\vartheta)P$ throughout the contact. At the ideal incoming crossing, $h = sy$. Set $A = Pb(j+1)$ and $a_0 = z_j/A$. At contact steps,

$$h_{j+n} = sy + (\vartheta - s)A(a_0 - T_n), \qquad T_n = \frac{H(j) - H(j+n)}{b(j+1)}, \quad 0 \leq T_n \leq n.$$

The scalar inputs and positions in this five-step window have magnitude less than $2A$, by the explicit formulas and perturbation bounds of Lemma 2. The invariant h lies between $sy - 5(\vartheta - s)A$ and $sy + (\vartheta - s)A$. Consequently every opposite-wall input and position is at least $sy - 7\vartheta A$; the same lower bound works for $w + z$. Since

$$\frac{y}{A} = \frac{\chi H}{b(j+1)} \geq \frac{j+3}{4},$$

these quantities are positive. Their sum also satisfies, for inputs as well as outputs,

$$w + z \leq sy + (2 - \vartheta)A < \left(.32 + \frac{24}{j+3}\right) y < 2.2.$$

Thus the other wall, the middle coordinates and the cap remain free. This validates the forced single-wall calculation as the actual recurrence. Before contact, both lower coordinates are nonnegative and their sum decreases, so no intermediate constraint has been missed.

Release is on an actual free interpolation interval between $j + N - 1$ and $j + N$, $N \in \{4, 5\}$. This gives the clock bound in (24). Conservation of the tangent momentum gives

$$p'_z = eP, \qquad p'_w = (s - \vartheta)P - \vartheta eP = -rP,$$

and therefore (23). The released coordinate increases strictly during the next free flight, excluding immediate recontact.

At the new section $z = 0$, so $D' = h$. The exact affine evolution of (26) proves (24). It also gives $0 < D' < .321y$. Since $\overline{H} \geq H(j+1)$,

$$\frac{|\delta_H|}{\overline{H}} \leq \frac{10}{j+1}.$$

Direct subtraction yields the exact identity

$$\chi' - \frac{s}{r}\frac{\chi}{1-\chi} = \frac{\delta_H}{rH'}\left(\vartheta - \frac{s}{1-\chi}\right).$$

The bracket has magnitude less than eight, and $r > .866$, proving (25). Finally,

$$\frac{.32}{.866} + .003 < .5, \qquad \frac{.1}{1.115}\,24 - .003 > .96.$$

These inequalities prove the two strict exit-face statements, with ample slack. ∎

7 One fixed parameter with infinitely many returns

At the first lower encounter, the two incoming wall momenta are both $-P_0$, and their ideal separation is 2ε. The same tangent coordinate proves isolation: throughout the contact, the inputs and positions of the other wall and the wall sum differ from 2ε by less than $12P_0b(j+1)$, whereas

$$\frac{\varepsilon}{P_0b(j+1)} > \frac{j+1}{2(1-c)} > 333333.$$

The scalar contact lemma applies. Put $R_1 = 1 + (1+e)\vartheta$. At the first outgoing section,

$$\begin{aligned} P_1 &= R_1P_0, & s_1 &= e/R_1, \\ D_1 &= 2\varepsilon + (1+\vartheta)P_0(H_1' - H_1). \end{aligned} \tag{27}$$

The relative clock error $\eta = (H_1' - H_1)/H_1$ has $|\eta| < 5 \cdot 10^{-6}$. Hence

$$\chi_1 = \frac{2/(1-c) + (1+\vartheta)\eta}{R_1(1+\eta)}. \tag{28}$$

It follows that $.103 < s_1 < .120$, $\kappa_1 > 10^6$, $0 < D_1 < 2.2$, and

$$\chi_1(-2) < .306 < .5, \qquad \chi_1(1/2) > 1.805 > .96. \tag{29}$$

Lemma 4. *The first outgoing section is continuous in c. Subsequent outgoing-section maps are continuous across all integer entry and release phase changes on realized block states.*

Proof. A realized section records its clock, its reduced position and its scaled momentum. These data recover the free pair at a subsequent integer N from $t_N = R - pH(N)$ and $t_N - t_{N-1} = pb(N)$, where $R = t(\kappa) + pH(\kappa)$. Locally one can choose a common N after release and before the next contact, because the release is transverse and the next encounter is strictly separated. Thus the section retains the information needed by the nonautonomous two-step recurrence.

Every finite ordinary trajectory is continuous in its initial data, because affine operations and coordinatewise clipping are continuous. Locally, compose ordinary updates to one fixed integer pair after all nearby releases and before any further encounter. Such a pair exists because the episode lasts at most five steps and the opposite-wall margin is strict. Recover the free scaled momentum there by dividing the increment by $b(N)$.

If the released coordinate at N is z_N with positive momentum p_z, its outgoing clock is $H' = H(N) + z_N/p_z$. The other section coordinates follow from the same affine free formula. All these operations are continuous with positive denominators.

At an entry-index transition, the appearing or disappearing leading clipping correction is zero; the two descriptions give the same ordinary states. At a release-index transition, the disputed correction is again zero and the crossing is at the common ordinary index. The neighboring descriptions have the same free scaled momentum because $b(k+1)/b(k) = \beta_k$. Thus both section descriptions have the same limit. The uniform positive release speed excludes tangential roots, and the isolation margins exclude simultaneous-wall ambiguities. These facts also hold in a small neighborhood of each particular realized block state. Every event under discussion occurs after finitely many ordinary steps, so finite trajectory continuity suffices. ∎

By (29) and Lemma 4, let c_b be the first value with $\chi_1(c_b) = .96$, and c_a the last value at or before c_b with $\chi_1(c_a) = .5$. On the compact interval $I = [c_a, c_b]$ the first section lies in the block. No monotonicity of the section map in c is assumed.

Let E_- and E_+ consist of parameters in I whose first strict exit is through the lower or upper χ face, respectively. All other coordinates remain in the block while χ does, by Lemma 3. The two exit sets are disjoint and relatively open. Indeed a trajectory first exiting below cannot have previously touched the upper face, since that would force an upper strict exit at the next return. By finite continuity, nearby parameters either exit below earlier or follow the same finite trajectory to a lower exit. The upper case is identical. This reasoning also covers an earlier touch of the eventual exit face.

The endpoints belong to opposite exit sets, because the face inequalities are strict. If every parameter eventually exited, the two nonempty relatively open sets would disconnect I. Therefore some

$$c_* \in I \setminus (E_- \cup E_+) \tag{30}$$

never exits. This is one fixed legal initialization. It is not a sequence of parameters retuned for successive finite horizons.

8 Divergent event times and primal pulses

For the selected orbit, (24) implies

$$.04 - 10^{-5} < H'/H < .5 + 10^{-5}.$$

Inverting (13), for $\kappa \geq 10^6$,

$$1.4 < \kappa'/\kappa < 5.1. \tag{31}$$

For example, $H(1.4\kappa)/H(\kappa) > .5 + 10^{-5}$ and $H(5.1\kappa)/H(\kappa) < .04 - 10^{-5}$; both are direct rational quadratic inequalities. Thus section times and ordinary contact indices

tend to infinity, with gaps at least proportional to current time. There is no finite-time accumulation, bounded-gap exponential capture, or eventually free termination.

Define $A_\ell = P_\ell b(\kappa_\ell)$. By (23) and (31),

$$\frac{A_{\ell+1}}{A_\ell} > \frac{.866}{(5.1)^3} > \frac{1}{200}. \tag{32}$$

At the second negative prediction of the next contact, the normalized value is

$$d_0(1+\beta_{j+1})(a_0-1) - \beta_{j+1}a_0.$$

Its magnitude is at least $\min\{d_0(1+\beta_{j+1}), \beta_{j+1}\} > .98$. The clipping correction at the actual update producing $m_\ell = j+2$ therefore has

$$\|\xi\|_1 \geq .98 P_\ell b(j+1).$$

Since $j+1 < \kappa_{\ell+1} < 5.1\kappa_\ell$, the exact primal certificate (12) yields

$$\|w_{m_\ell} - \mathbf{1}\| > \frac{.98}{\sqrt{11}(5.1)^3} A_\ell > \frac{A_\ell}{10000}. \tag{33}$$

These are ordinary integer iterates, not interpolated points or merely dual displacements.

Finally, $\kappa_\ell \geq \kappa_1 (1.4)^{\ell-1}$ and $A_\ell > A_1 200^{-(\ell-1)}$. With q in (6),

$$A_\ell \geq A_1 \kappa_1^q \kappa_\ell^{-q} \geq A_1 \kappa_1^q m_\ell^{-q},$$

because the contact occurs after the current section. Equation (33) proves the endpoint assertion of Theorem 1. For every $\rho \in (0,1)$, $m^{-q}/\rho^m \to \infty$, so the lower bound excludes every eventual R-linear estimate.

The distances at outgoing sections decrease by a factor below .321. Since $A_\ell / y_\ell = 2/(\chi_\ell \kappa_\ell) \leq 4/\kappa_\ell$, the contact excursions vanish together with the section distances. The free segments join points in the same shrinking neighborhood. Consequently the dual orbit approaches the apex, where the positive multiplier is $(0, \ldots, 0, 11)$. All primal margins remain active at the limit, and ten multipliers vanish. This is precisely the degenerate limiting geometry used by the construction. The strictly positive optimal multiplier $\mathbf{1}$ remains available elsewhere in the solution triangle.

9 The fixed strict step and its full state

We now fix

$$\delta = 10^{-30}, \qquad \gamma = 1 - \delta. \tag{34}$$

All data, the box schedule and the initialization are those of (3)–(5); in particular the term $38\delta/9$ in Y_0 is retained. The global bound (10) already proves that the boxes are inactive and that $w_k \to \mathbf{1}$. What remains is the lower subsequence. Finite-horizon continuity in

γ would not establish this assertion: range memory must be controlled through infinitely many contacts at this one fixed positive δ.

Write

$$t_k = B^T v_k, \qquad r_k = Q v_k - \mathbf{1}, \qquad v_k = \mathbf{1} + B t_k + r_k.$$

A hat denotes extrapolation with $\beta_k = k/(k+3)$, for example $\widehat{r}_k = (1+\beta_k) r_k - \beta_k r_{k-1}$. The exact unboxed iteration is

$$\begin{aligned} h_{k+1} &= \mathbf{1} + B\widehat{t}_k + \delta \widehat{r}_k, \quad \xi_{k+1} = [-h_{k+1}]_+ , \\ t_{k+1} &= \widehat{t}_k + B^T \xi_{k+1}, \qquad r_{k+1} = \delta \widehat{r}_k + Q\xi_{k+1}, \quad w_k - \mathbf{1} = r_k. \end{aligned} \tag{35}$$

Here h_{k+1} is the complete clipping input, whereas $\mathbf{1} + B\widehat{t}_k$ is only its null-coordinate part. Their distinction matters even when a clipping amount is arbitrarily small.

On genuinely free updates the null clock (14) is unchanged. Set

$$M_k = \max\{\|r_k\|, \|r_{k-1}\|\}.$$

Since $0 < \beta_k < 1$ and $3\delta < 1$, the exact range recurrence gives

$$\|r_{k+1}\| \le 3\delta M_k, \quad M_{k+1} \le M_k, \qquad M_{k+2} \le 3\delta M_k \tag{36}$$

when both updates in the last inequality are free. These estimates will be compared with the current contact scale, rather than a fixed absolute tolerance.

9.1 The cap launch with the modified initialization

At output two the input is $(1+19\delta)\mathbf{1} + Bt_0$. Only the cap is negative, with clipping amount

$$\ell = 10Y_0 - 11 - 19\delta > 0.$$

The identities $Qe_{11} = \mathbf{1}/11$ and $bB_{11,2} = -1/11$ give

$$\begin{aligned} X_2 &= \varepsilon, & Y_2 &= 1 + Y_0/11 + 19\delta/11, \\ r_2 &= \eta_2 \mathbf{1}, & \eta_2 &= 19\delta + \ell/11, \\ Y_3 &= 11/15 + 19\delta/15 + c\varepsilon/3, & P_0 &= 60(Y_2 - Y_3) = \frac{88 + 152\delta - 80c\varepsilon}{3}. \end{aligned} \tag{37}$$

The next update is strictly free: the lower null coordinates are close to $11/15$, the cap is greater than 3.6, and the range contribution is of order δ. More precisely its uniform range coefficient is

$$\eta_3 = \frac{\delta}{9}(-53 + 266\delta - 14c\varepsilon). \tag{38}$$

The subsequent free null line is therefore

$$X_k = \varepsilon, \qquad Y_k = c\varepsilon + P_0 H(k), \qquad 29 < P_0 < 30. \tag{39}$$

The constant $c\varepsilon$ follows from $2Y_2-Y_0=c\varepsilon$, so no step-dependent drift has been suppressed.

Define the ideal target clock and its integer bracket by

$$H_c=\frac{(1-c)\varepsilon}{P_0},\qquad H(j)\ge H_c>H(j+1),\qquad 2\cdot 10^6<j<6\cdot 10^6. \tag{40}$$

The last bounds follow by substituting $c\in[-2,1/2]$ and $29<P_0<30$ into H. This j is an ideal null-line index. It need not be an actual last-free index.

We verify the free flight through output $j-1$. Initially $M_2<64$ and, by (38), $M_3<5$. Through every output $n\le j-1$ after output three, the target value on (39) is at least $P_0b(j)>10^{-19}$. The other lower value is larger by 2ε, the middle values are their mean, and the cap keeps positive slack because Y_n decreases from Y_3. Under the free induction the complete range input has norm at most $15\delta<10^{-28}$. Thus all these updates are indeed free. After the six free updates from output four through output nine, (36) yields

$$M_9<135\delta^3<1.4\cdot 10^{-88}.$$

Since $P_0b(j+1)>10^{-19}$, monotonicity on the rest of the flight proves

$$M_{j-1}<\delta^2P_0b(j+1). \tag{41}$$

Output j may already clip weakly if its null input is sufficiently close to zero. The contact analysis below begins from the verified free pair at $j-2,j-1$ and includes that possibility.

10 A uniform contact bound with memory

We use $g=28/55$, $d_0=27/55$ and $\vartheta=27/28$ as above. Consider an incoming target line

$$y_k=P(H(k)-H_c),\qquad P>0,$$

with $H(j)\ge H_c>H(j+1)$ and $j\ge 10^6$. Put

$$A=Pb(j+1),\qquad h=\frac{H(j)-H_c}{b(j+1)}\in[0,1). \tag{42}$$

Suppose the actual null pair at $j-2,j-1$ lies on this line and $M_{j-1}\le 134A$. Temporarily force only the target coordinate to clip in (35). We will verify all other inputs when applying this calculation.

Lemma 5. *For this forced contact, every newly generated range vector through output $j+6$ has norm less than $5A$, and $M_k\le 134A$ throughout the window. The complete target inputs at $j+2$ and $j+3$ are below $-.97A$ and $-.46A$, respectively. The updates at $j+5,j+6$ are free, with target inputs greater than $.017A$ and $.25A$. Moreover*

$$\begin{gathered}.017<y_{j+5}/A<.268,\qquad .2405<(y_{j+6}-y_{j+5})/A<.2501,\\ e:=\frac{y_{j+6}-y_{j+5}}{Pb(j+6)}\in(.23,.26),\qquad M_{j+6}\le 15\delta A.\end{gathered} \tag{43}$$

These bounds include all incoming phases and a possible weak contact at output j.

Proof. First set $\delta = 0$ while keeping the incoming null pair and the actual coefficients β_k. At outputs j and $j+1$ the normalized target values are exactly h and $d_0(h-1)$. The one-wall scalar map is ψ from (18), which is one-Lipschitz. Freeze the subsequent coefficients at one. The predictors at offsets two, three and four are

$$-\frac{h+54}{55}, \qquad -\frac{1539h+1431}{3025}, \qquad \frac{2916-81621h}{166375}.$$

Only the last expression changes sign, at $h = 108/3023$. Continue both branches, using the identity branch when that expression is nonnegative and multiplication by d_0 otherwise. Offsets five and six are free in both branches. If u_n denotes this frozen sequence, the exact extrema are

$$\begin{aligned} \frac{2916}{166375} \le u_5 \le \frac{44469}{166375}, \\ \frac{729}{3023} \le u_6 - u_5 \le \frac{41553}{166375}, \quad u_6 > .26. \end{aligned} \tag{44}$$

These follow by evaluating the affine expressions on the two phase intervals; the relevant offset-four and velocity expressions are already displayed in the proof of Lemma 2. Every consecutive frozen increment through offset six has absolute value at most one.

Let $\eta = 3/j$. Restoring the true β_k perturbs the output and predictor at offsets two through six by at most

$$\eta, \quad 3\eta, \quad 8\eta, \quad 20\eta, \quad 49\eta,$$

respectively. Indeed, each new bound is twice the preceding bound plus the one before it plus η, starting from two zeros. The error in the final velocity is at most 69η. Substitution in (44), with $j \ge 10^6$, proves the stated position, velocity and threshold bounds with strict slack for the endpoint forced recurrence. In particular its negative threshold magnitudes are less than $1.001A$.

Now restore the fixed δ. For a target row B_i, the endpoint null map is

$$T_i(t) = t + B_i^T[-(1 + B_i t)]_+.$$

Its derivative on the clipped branch is $I - B_i^T B_i$, whose eigenvalues are 1 and $1 - g$, and on the other branch it is the identity. Thus T_i is nonexpansive. A range input changes this map by at most $\|\delta \widehat{r}_k\|$. Under the provisional bound $M_k \le 134A$, this forcing is at most $402\delta A$. Therefore successive full null-state errors, and the corresponding complete target-input errors, satisfy the majorant recurrence

$$E_{n+1} \le 2E_n + E_{n-1} + 402\delta A.$$

Starting with the identical pair at $j-2, j-1$, the seven forcing multipliers for outputs $j, \dots, j+6$ are

$$1, \quad 3, \quad 8, \quad 20, \quad 49, \quad 119, \quad 288.$$

Every comparison error is consequently at most

$$288 \cdot 402\delta A < 10^{-24} A. \tag{45}$$

This also closes the provisional bound. At each step its correction is less than $1.002A$, so (35) produces a new range vector of norm at most $402\delta A + 1.002A < 5A$. Induction preserves $M_k \le 134A$, including the two incoming states. No constant in this argument depends on P, A or the number of preceding contacts.

The strict threshold margins at $j+5, j+6$ make these two updates genuinely free for the forced recurrence. At $j+4$ both range states have norm below $5A$, so (36) gives $M_{j+6} \le 15\delta A$. Finally

$$1 < \frac{b(j+1)}{b(j+6)} < 1 + \frac{16}{j}$$

converts the physical velocity bound into $.23 < e < .26$. The comparison began before output j, so an additional vanishing or weak initial clipping has not been omitted. ∎

10.1 An extrapolated release section

Extend the exact null free line determined by the actual pair at $j+5, j+6$ backwards, and define

$$H' = H(j+5) + \frac{y_{j+5}}{eP}, \qquad \kappa' = H^{-1}(H'), \qquad \Delta = H_c - H'. \tag{46}$$

This is a zero of the outgoing null line; it need not be a zero on an actual free update. This distinction replaces the endpoint crossing convention.

The extended value at $j+3$ is

$$y_{j+5} - \frac{y_{j+6} - y_{j+5}}{b(j+6)}\big(b(j+4) + b(j+5)\big) < (.268 - 2 \cdot .2405)A < 0.$$

Its value at $j+5$ is positive. Hence

$$j+3 < \kappa' < j+5, \qquad 0 < \Delta \le 5b(j+1), \qquad \frac{\Delta}{H_c} < \frac{10}{j+1}. \tag{47}$$

The last two inequalities use $H(j) \ge H_c > H(j+1)$ and the telescoping clock identity. The section will retain the two actual range states at a later free integer pair. Defining the extrapolated zero does not restart either range state.

11 An invariant block retaining both range states

Orient an outgoing extrapolated section so that the released null wall is x. Its free null line is specified by

$$x(\kappa) = 0, \quad y(\kappa) = D = \chi H P, \quad p_x = sP, \quad p_y = -P, \quad H = H(\kappa). \tag{48}$$

The numerical block is

$$\kappa \geq 10^{6}, \quad .1 \leq s \leq .32, \quad .5 \leq \chi \leq .96, \quad 0 < D < .1, \quad P > 0. \tag{49}$$

At the canonical integer $N = \lfloor \kappa \rfloor + 10$, the section also carries the actual pair r_N, r_{N-1}, the null pair on the line (48), and the requirement

$$M_N \leq \frac{Pb(\kappa)}{100}. \tag{50}$$

Both states in this pair are after the preceding episode, on its genuinely free continuation. Equivalently, one can retain the full free range solution and represent it by a different pair of integer values. Such a change of coordinates does not alter the physical trajectory.

Lemma 6. *Every realized full section satisfying* (48)–(50) *has an isolated opposite-wall encounter and a subsequent section. All conditions except the two bounds on* χ *are p-reserved with strict slack. More precisely, with* j *its next ideal incoming index, the next section satisfies*

$$\begin{aligned} R &= (1+e)\vartheta - s, \qquad P' = RP, \qquad\qquad s' = e/R, \\ H' &= (1-\chi)H - \Delta, \quad D' = sD - (\vartheta - s)P\Delta, \end{aligned} \tag{51}$$

where e *and* Δ *obey* (43) *and* (47)*. In addition,*

$$\begin{gathered} .866 < R < 1.115, \qquad .206 < s' < .301, \qquad 0 < D' < .32D, \\ 1.4 < \kappa'/\kappa < 5.1, \qquad \left| \chi' - \frac{s}{R}\frac{\chi}{1-\chi} \right| < \frac{82}{j}, \\ M_{N'} \leq 15\delta P b(j+1) < 18\delta P' b(\kappa') < P' b(\kappa')/100, \qquad N' = \lfloor \kappa' \rfloor + 10. \end{gathered} \tag{52}$$

The face $\chi = .5$ *maps strictly below* .5*, and the face* $\chi = .96$ *strictly above* .96*.*

Proof. The free null lines are

$$x_k = sP[H - H(k)], \qquad y_k = D - P[H - H(k)]. \tag{53}$$

The next target root has clock $H_c = (1-\chi)H$. The exact inverse

$$H^{-1}(u) = -\frac{3}{2} + \sqrt{\frac{1}{4} + \frac{1}{2u}}$$

and the bracket $H(j) \geq H_c > H(j+1)$ give

$$1.4\kappa < j + 1 < 5.1\kappa, \qquad j - 1 > N, \qquad b(\kappa) < 134\, b(j+1). \tag{54}$$

For instance the upper estimate follows first as $j + 1 < 5\kappa + 7$; substituting this into the product defining b gives the last bound for $\kappa \geq 10^{6}$. The lower bound follows by comparing $H(1.4\kappa)$ with $H(\kappa)/2$.

We first justify the free flight. Direct substitution in (53), using $\kappa+8 < N-1 \le \kappa+9$, gives

$$x_{N-1} > .79Pb(\kappa), \qquad x_N > .89Pb(\kappa).$$

Also $y_N > D - 10Pb(\kappa) > (1 - 4 \cdot 10^{-5})D$, since $Pb(\kappa)/D = 2/(\chi\kappa)$. The opposite wall therefore dominates the allowed range error at the canonical pair. The sum $x + y$ is positive and decreasing along this flight, with $x + y \le D < .1$. Thus the canonical full pair is feasible under (50). In particular the cap has positive slack and the middle coordinates are positive. At each subsequent proposed output through $j - 1$ the target null value is at least $Pb(j) > A$, where $A = Pb(j + 1)$, and the released wall increases. Under the free induction (36) and (50) bound every range contribution to an input by

$$.03\delta Pb(\kappa) < 4.02\delta A.$$

This is smaller than all the foregoing positive margins. Every such output is therefore genuinely free, proving the induction and the hypotheses of Lemma 5 at the pair $j-2, j-1$.

It remains to verify that the forced target calculation is the actual eleven-dimensional recurrence. During isolated target clipping the combination

$$L_k = x_k + \vartheta y_k$$

is unaffected by the impulse, because the off-diagonal lower-wall Gram entry is $-\vartheta$ times the diagonal entry. Its scaled momentum is exactly $(s - \vartheta)P$, even in the presence of range memory. At the ideal target root its value is sD. From (54),

$$\frac{D}{A} = \frac{\chi H(\kappa)}{b(j + 1)} > .68\kappa \ge 680000. \tag{55}$$

Over the seven-output window, both the null target values and their null predictors have magnitude below $2A$, by the proof of Lemma 5. The tangent coordinate and its predictors differ from sD by less than $8A$. Hence the opposite null input and the null wall sum are greater than $.1D - 10A > 0$. Their upper bounds are less than $D + 20A < .101$. The corresponding complete inputs add only $\delta\widehat{r}$, of norm at most $402\delta A$. Thus the opposite wall and every middle coordinate remain strictly positive, while the cap input exceeds 10.49. No non-target clipping occurs. The forced recurrence, including the allowed weak contact at j, is the actual recurrence.

After outputs $j + 5, j + 6$, the released target is positive and increases on its free line. The other wall has margin at least $.1D - 10A$. Through output $j + 15$ it can lose less than a further $20A$, which is still negligible relative to (55). The middle and cap inputs keep their strict margins, and the range is bounded by $15\delta A$ on this free continuation. This excludes immediate recontact and verifies a genuinely free segment through

$$N' = \lfloor \kappa' \rfloor + 10 \in \{j + 13, j + 14\}.$$

It also provides common free pairs on either side of an integer change in κ'.

Conservation of the tangent momentum gives $p'_y = eP$ and $p'_x = -[(1+e)\vartheta - s]P$. Interchanging the wall names gives the first line of (51). Evaluating the same tangent line at $H' = H_c - \Delta$ gives its second line. The bounds on R and s' follow by substitution of $.23 < e < .26$ and $.1 \le s \le .32$. Since $\vartheta > s$, $\Delta > 0$ and $P\Delta \le 5A$, (55) gives

$$0 < D' = sD - (\vartheta - s)P\Delta < sD \le .32D.$$

Also $H'/H = (1-\chi)(1-z)$, where $z = \Delta/H_c < 10/(j+1)$. Comparing the explicit clock at 1.4κ and 5.1κ proves the time-ratio bounds in (52).

For the normalized distance, direct subtraction in the exact identities yields

$$\chi' - \frac{s}{R}\frac{\chi}{1-\chi} = \frac{z}{R(1-z)}\left(\frac{s}{1-\chi} - \vartheta\right). \tag{56}$$

The absolute value of the bracket is below 7.04, $R > .866$ and $z < 10/(j+1)$, so the error is below $82/j$. The strict face inequalities follow from

$$\frac{.32}{.866} + .003 < .5, \qquad \frac{24 \cdot .1}{1.115} - .003 > .96.$$

Finally $\kappa' < j + 5$ and $R > .866$ imply

$$P'b(\kappa') > .866Pb(j+5) > .85A.$$

By the two free release updates and the verified continuation to N', we obtain

$$M_{N'} \le 15\delta A < 18\delta P'b(\kappa') < P'b(\kappa')/100.$$

This is a relative bound at the new scale. It remains valid at every return even as that scale tends to zero, and it uses the actual range vectors generated by the original iteration. ∎

12 A fixed parameter and primal lower bounds

12.1 The first full section

For the initial flight (39), both incoming lower-wall momenta are $-P_0$ and their separation at the ideal target root is 2ε. With $A = P_0 b(j+1)$, (40) gives

$$\frac{\varepsilon}{A} > \frac{j+1}{2(1-c)} > 333333.$$

The tangent coordinate $x + \vartheta y$ is 2ε at that root. During the first seven-update window and the free continuation through $j+15$, the opposite-wall and middle inputs differ from their positive null values at the root by less than $40A$. The cap remains far from zero. Together with (41), these inequalities verify isolation for this first encounter by the same forced calculation. The first complete section is therefore defined for every $c \in [-2, 1/2]$.

Set $R_0 = 1 + (1+e)\vartheta$. After interchanging the wall orientation, the exact first-section identities are

$$\begin{aligned} P_1 &= R_0 P_0, \qquad s_1 = e/R_0, \\ H_1 &= H_c - \Delta, \quad D_1 = 2\varepsilon - (1+\vartheta)P_0\Delta. \end{aligned} \tag{57}$$

In particular $2.186 < R_0 < 2.215$, $.103 < s_1 < .120$, $\kappa_1 > 2 \cdot 10^6$ and $0 < D_1 < 2\varepsilon < .1$. For $z = \Delta/H_c < 5 \cdot 10^{-6}$,

$$\chi_1(c) = \frac{2/(1-c) - (1+\vartheta)z}{R_0(1-z)}. \tag{58}$$

The signs of the z corrections at the endpoint parameters show directly that

$$\chi_1(-2) < \frac{2}{3R_0} < .306, \qquad \chi_1(1/2) > \frac{4}{R_0} > 1.805.$$

The two free release updates give the canonical memory inequality with the stronger slack in (52). These estimates establish every first-section condition except the desired restriction on χ_1, which will be imposed by choosing the parameter interval.

12.2 Continuity without resetting range memory

Every finite full-state trajectory is continuous in c, because each ordinary step consists of fixed affine maps followed by coordinatewise clipping. To transfer this fact to sections, fix one realized finite history and choose a common integer pair after its release and before its next encounter. The strict free margins proved above ensure that the same pair remains free for all nearby parameters. In particular this is possible when the ideal j changes by one, a weak early clipping appears, or a later clipping amount vanishes: the seven-update estimates hold on every phase, and the free segment extends through $j+15$.

At this fixed pair, recover the null momentum by dividing its increment by the positive clock difference $b(N)$. The released momentum is nonzero. Its line then determines H', κ', D', P', s' and χ' continuously, using (46). Retain the two range vectors at this same pair, so the complete future state is also retained. Recovering the line from another pair on the same free segment gives identical section quantities.

When κ' crosses an integer, the canonical index $\lfloor \kappa' \rfloor + 10$ changes. Its two range vectors are coordinate representatives of the same free range solution; no assertion of global Euclidean continuity of this floor-dependent representation is needed. Comparisons are made at the common fixed pair just described. The memory inequality holds with the strict bound $18\delta P' b(\kappa')$ at all these nearby pairs. Thus canonical changes cannot cause an unrecorded memory exit. This establishes continuous local extensions of every finite sequence of section functions of the original parameter c.

12.3 Nested parameter intervals

By (58) and continuity, there is a closed interval $I_1 \subset [-2, 1/2]$ on which $\chi_1 \in [.5, .96]$, with lower and upper face values at its two endpoints in this order. One construction

takes the first upper-face crossing and the last lower-face crossing before it. All the other section conditions have already been proved with strict slack.

Suppose a nonempty closed interval I_n has all its first n section histories in the block, and its nth normalized distance takes the two face values at the endpoints. Lemma 6 constructs the next section for every parameter in I_n, preserves every other condition including the complete memory bound, and gives a continuous next normalized distance. At the two endpoints this next distance is strictly below .5 and strictly above .96, respectively. Taking the first upper-face crossing and the last preceding lower-face crossing again produces a nonempty closed subinterval $I_{n+1} \subset I_n$ on which all $n+1$ histories stay in the block, with the required new endpoint values.

Nested compactness gives

$$c_* \in \bigcap_{n \geq 1} I_n. \tag{59}$$

For this one parameter, the original iteration has infinitely many admissible full-state returns. The step defect remains the single rational value (34). Neither the initialization nor the step is retuned at successive horizons.

12.4 Primal pulses after range cancellation

For the selected orbit, put $A_\ell = P_\ell b(\kappa_\ell)$. The return bounds imply

$$\frac{A_{\ell+1}}{A_\ell} = R_\ell \frac{b(\kappa_{\ell+1})}{b(\kappa_\ell)} > \frac{.866}{5.1^3} > \frac{1}{200}. \tag{60}$$

Take the ordinary output $m_\ell = j_\ell + 2$. The exact release bounds ensure

$$\kappa_\ell < m_\ell < \kappa_{\ell+1}.$$

At this output only the contacting wall clips. If its index is $i \in \{1, 2\}$, Lemma 5 gives

$$\xi_{m_\ell} = \xi_i e_i, \qquad \xi_i > .97 P_\ell b(j_\ell + 1), \qquad \|\delta \widehat{r}_{m_\ell - 1}\| \leq 402 \delta P_\ell b(j_\ell + 1).$$

Here $\|Qe_i\| = \sqrt{27/55} > 1/\sqrt{11}$. Thus the actual primal residual in (35) obeys

$$\begin{aligned} \|w_{m_\ell} - \mathbf{1}\| &\geq \|Qe_i\|\, \xi_i - \|\delta \widehat{r}_{m_\ell - 1}\| \\ &> \left(\frac{.97}{\sqrt{11}} - 402\delta \right) P_\ell b(j_\ell + 1) > \frac{A_\ell}{10000}. \end{aligned} \tag{61}$$

The final inequality uses $b(\kappa_\ell) < 134 b(j_\ell + 1)$. Possible cancellation by the retained range component has therefore been included, rather than inferred from a null displacement.

Finally $\kappa_\ell \geq \kappa_1 (1.4)^{\ell - 1}$ and $A_\ell > A_1 200^{-(\ell-1)}$. With $q = \log 200 / \log 1.4$,

$$\|w_{m_\ell} - \mathbf{1}\| > \frac{A_1 \kappa_1^q}{10000}\, m_\ell^{-q}.$$

The indices diverge, so this excludes every eventual R-linear estimate. Primal convergence was supplied independently by the boxed energy, completing the strict-step rate assertion.

For completeness, $D_{\ell+1} < .32 D_\ell$ and

$$\frac{A_\ell}{D_\ell} = \frac{2}{\chi_\ell \kappa_\ell} \longrightarrow 0.$$

The null coordinates between successive sections are $O(D_\ell)$, by the free lines and the uniform contact estimates. The range states are $O(A_\ell)$ throughout the contact windows and their intervening free flights. Hence $v_k \to 11 e_{11}$, proving the remaining dual-limit assertion of Theorem 1.

The theorem concerns two specified steps and equal nonzero initialization. It does not assert the same behavior for every strict step, typical initializations, or the zero-vector start. In fact at $\delta = 0$ this projector instance reaches the primal optimum immediately from the zero-vector start. The selected parameters are given by the nonexit construction (30) and the nested intervals (59); no decimal shooting approximation is required. The exponent is a quantitative obstruction to a geometric rate, not a sharp asymptotic claim.

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